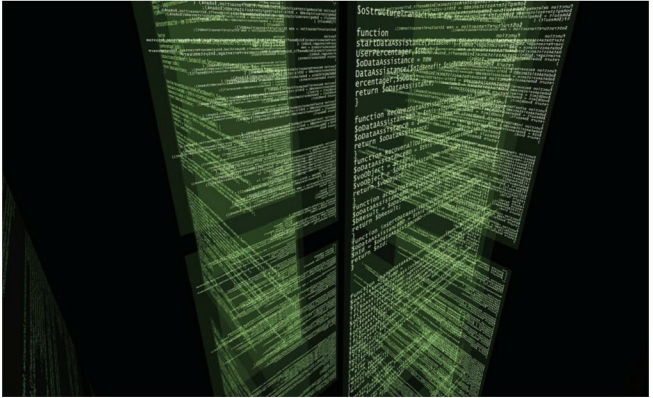


Scilab: The Open Source Equivalent of MATLAB

Scilab is free and open source software for numerical computation. It provides engineers and scientists with a powerful environment to do their computation.



Engineers and scientists all over the world use computers for their calculations, yet many of us remain unaware of the software or tools used by a majority of them. Since the invention of transistor in 1947, computing power has increased tremendously, more or less according to the predictions made in Moore's Law. The law states that the number of transistors in a given area will be doubled approximately every two years. In 2015, the transistor count was 5.5 billion in an Intel processor. That is around 5.5 per cent of the 100 billion neurons in the human brain. So, artificial intelligence (AI) is getting closer to human intelligence.

MATLAB is a high level numerical computational platform with IEEE floating point precision of approximately 16 decimal digits. However, space exploration demands a greater level of decimal precision—approximately 25 digits is the count for ISRO. So do NASA and ISRO use MATLAB? Yes, they do, for most of their computation. But they also use Scilab, an open source, high level numerical computational platform.

Scilab can be downloaded from www.scilab.org. Taking up about 150MB, it can be installed on your Linux, Windows or Mac OS. Learning Scilab is easy with <http://www.scilab.in> and with the initiatives taken by Prof. Kannan M. Moudgalya at the Indian Institute of Technology, Bombay. The professor's textbook companion project at www.scilab.in/Completed_Books contains almost all the engineering numerical problems solved with Scilab. These can be accessed freely, but sadly, most engineering students are unaware of this.

MATLAB

Cleve Moler, the chairman of the computer science department of the University of New Mexico, started developing MATLAB in the late 1970s. The latest release of this proprietary programming language has a memory size of more than 7GB. It has many inbuilt functions and simulation tool bars, which make it easy to compute on various platforms according to a user's needs. It has well written help documentation and wide online community support. For